

The Sound Of Silence - Final Report

Group 9

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1 Introduction

Silences are an important part of human dialogue [2]. They provide crucial information about the mood and intention of a dialogue which most humans can easily pick up on. However, given their implicit, non-linguistic nature, they are not well modeled in NLP systems. Despite their communicative significance, little work examines how Large Language Models (LLMs) interpret them. This project explores how LLMs interpret silences in dialogue and whether their classifications of the said silences into categories align with human annotations of the same silences.

To evaluate how LLMs handle silences, we performed a short proof-of-concept. We took multiple examples of audio with meaningful silences in the middle and prompted models to classify their intended meaning. We presented each model with two different modalities: one time, a transcribed audio clip with a marked silence token in the middle, and the second time we send both the transcript and the original audio clip. This experimental setup allows us to compare the contribution of textual versus audio information and to assess how each modality influences the model's interpretation of silence.

LLM Prompt Example

You are given a short dialogue segment with a silence between two speakers. Your task is to classify the function of the silence using one of the following labels:

- PRAGMATIC
- RETRIEVAL
- PROCESSING
- EMOTIONAL
- SURPRISE
- EMNARRASSMENT

Return only the label, with no explanation.

The dialogue:

A: *when the cameras are rolling and the 20,000 fans*

are there and the cheerleaders are dancing and the popcorn is popping and the celebrities are sitting in Celebrity row but like who has the love for the game when nobody is there

[SILENCE: 2.10 seconds]

B: *I love all of that and it's interesting the Love of the Game part is so funny to me because I think there's a distinction between loving the game the actual competition itself and what it means to actually love the game and everything that goes into that.*

GPT Output: PROCESSING

Correct Label: PRAGMATIC

This demonstrates that the model misclassified a short natural pause as a thinking related silence and reflects model performance gap

Preliminary results on this POC show that LLMs often miss subtle conversational cues, highlighting the need for human annotation. Therefore, in this work, we construct a pipeline for the annotation of training data to improve LLMs' understanding of the complex concept of silence. We construct the first ever annotated dataset of conversational silences, derived from two-speaker podcasts in Hebrew and labeled according to six pragmatic silence categories. To create the annotations, the data went through several rounds of annotations by several groups of people.

2 Related Work

The study of silence in human communication spans linguistic theory, pragmatics, and signal processing. While often perceived merely as the absence of sound, recent scholarship emphasizes that silence is an active component of dialogue. Khademi [4] notes that silence is "multi-deterministic," deriving specific meanings—such as agreement, disagreement, or hesitation, entirely from the sequential context of the conversation. Similarly, López Gutiérrez et al. [3] argue that silence is a potent tool in communication, capable

of expressing complex states such as respect, empathy, or fear, and differentiate between "objective silence" (physical absence of noise) and "subjective silence" (intentional communicative acts). Dupuche [2] posits that silence is an "essential aspect of dialogue," facilitating deep listening and authentic exchange. Scholars have developed various typologies to categorize these functions. Kurzon [5] distinguishes between "intentional silence" (strategic decision not to speak) and "unintentional silence" (caused by psychological inhibition or cognitive processing). Expanding on the relational impact of these pauses, Wu et al. [6] apply Rapport Management Theory to categorize silence into four pragmatic orientations: maintaining, neglecting, challenging, or destroying interpersonal harmony. They argue that silence is not a passive withdrawal but a dynamic non-verbal strategy used for conflict avoidance, emotional regulation, or aggression. From a technical perspective, Atal et al. [1] established pattern recognition approaches to classify speech segments into voiced, unvoiced, and silence categories using acoustic measurements, demonstrating early computational approaches to silence detection. Despite this theoretical foundation, no existing dataset provides aligned audio and text with annotated conversational silences for computational analysis. This gap limits our ability to evaluate how LLMs interpret these non-linguistic events. Our work addresses this by constructing an annotated silence dataset with pragmatic categories, enabling systematic evaluation of whether LLMs can capture the subtle communicative functions that human annotators identify in both textual and auditory contexts.

3 Data Acquisition

This section outlines the workflow used to collect, process, and prepare the audio data for silence detection and annotation. The pipeline includes sourcing audio from Hebrew-language podcasts, transcribing and diarizing speech with AssemblyAI¹, computing dynamic silence thresholds, extracting aligned text and audio segments, and performing quality control to ensure dataset integrity.

3.1 Data Sources

We collected conversational audio from two categories of publicly available material:

1. Two-speaker podcast episodes obtained from Spotify.
2. Short dialogue scenes from movies and television series downloaded from YouTube.

All recordings were converted to **MP3 format** to standardize downstream processing. These sources provide structured dialogue, recurring speakers, and relatively stable acoustic conditions, making them appropriate for timing-based analyses of silence.

3.2 Transcription and Speaker Diarization

Each MP3 file was processed using the AssemblyAI API, which performs automatic speech-to-text transcription and speaker diarization. The diarization step produces a chronological sequence of utterances, where each utterance is annotated with a **speaker identifier** (e.g., Speaker A or Speaker B), **start and end timestamps**, and the corresponding **transcript text**. An example of the resulting structure is shown below (simplified):

Speaker A | 12.0s-15.8s | "I think the solution is simple." Speaker B | 17.1s-19.4s | "Why do you think so?"

3.3 Silence Detection and Context Extraction

From the diarized transcripts, we computed the temporal gap between each pair of consecutive utterances by subtracting the end time of Speaker A's segment from the start time of Speaker B's segment. For example, if an utterance ends at 15.8 s and the next begins at 17.1 s, the resulting gap is 1.3 s.

To identify meaningful pauses, we used a dynamic threshold that adapts to each audio file. We computed the distribution of all inter-speaker gaps and defined a silence interval as any gap above the 75th percentile and lasting at least 2 seconds. This selects the longest pauses while accounting for each recording's pacing.

For every detected silence, we extracted its conversational context by taking the three utterances before the silence and the three after it. These six utterances were concatenated into a single `combined_text` field, providing a localized context window for downstream analysis.

3.4 Audio Clip Generation and Final Dataset Construction

To refine the dataset, the research team manually filtered low-quality audio by removing advertisements and visual-only references to isolate genuine

¹<https://www.assemblyai.com/>

conversational silences. For each detected interval, we extracted segments from the original MP3 files and saved them individually to be compiled into a structured dataset. Within this dataset, each row includes the audio path and silence duration along with speaker information, surrounding utterances, and a six-utterance context window. The dataset is public and available in Hugging Face².

3.5 Silence Labels Definition

We define six labels that capture different pragmatic roles of silence:

1. **Pragmatic Silence** – Silence that fulfills a communicative or social role in the conversation. It serves as a deliberate cue, such as signaling a topic shift, guiding the conversation in a specific direction, or encouraging the interviewee to elaborate.
2. **Retrieval Silence** – A brief pause occurring when the speaker knows what they want to convey but stops momentarily to retrieve specific information, refine their phrasing, or organize a simple sentence.
3. **Processing Silence** – A pause occurring when the speaker deeply processes what has been said to formulate a stance, consider broader meanings, or construct a complex response. Its role is to organize thoughts for a significant conversational move, such as an interpretation, a challenge, or a complex argument.
4. **Emotional Silence** – Silence occurring when emotional weight makes it difficult for a speaker to continue, such as during moments of sadness, excitement, or stress. It can also occur when a listener is moved by what is said and pauses to process the feeling.
5. **Surprise Silence** – A pause triggered by an unexpected event, statement, or behavior that deviates from a person’s known persona or the conversation’s expectations. This silence marks a momentary stay to digest the unexpected information.
6. **Embarrassment Silence** – Silence caused by an unexpected remark or behavior that potentially violates social norms, creating social

²https://huggingface.co/datasets/Zivf/NLP_The_Sound_Of_Silence_Datasets

Label	Cohen’s κ
PRAGMATIC	0.721
RETRIEVAL	0.704
PROCESSING	0.660
EMOTIONAL	0.700
SURPRISE	0.519
EMBARRASSMENT	0.293
Overall	0.677

Table 1: Inter-annotator agreement measured using Cohen’s Kappa.

Annotator Pair	Cohen’s κ
Annotator A – Annotator B	0.747
Annotator A – Annotator C	0.725
Annotator B – Annotator D	0.632
Annotator C – Annotator D	0.598

Table 2: Pairwise inter-annotator agreement measured using Cohen’s Kappa.

discomfort. It represents a momentary awkwardness where the speaker is searching for how to react to the situation.

3.6 Internal Team Annotation

The primary goal of this phase was to generate a ground-truth dataset, which serves two critical purposes: first, as a gold standard to evaluate the performance of LLMs on the silence classification task, and second, as a benchmark to analyze and validate the reliability of results from external annotators. To this end, we conducted an internal annotation process. Each team member was randomly assigned samples from the dataset and annotated them according to the established label definitions.

The overall Cohen’s Kappa score across all labels was $\kappa = 0.677$, indicating substantial agreement according to standard interpretation guidelines. Table 1 reports per-label Kappa scores, revealing variation in agreement across silence types.

In addition to the overall agreement score, we computed pairwise Cohen’s Kappa values between all annotator pairs. Pairwise agreement ranged between $\kappa = 0.598$ and $\kappa = 0.747$, indicating consistent annotation behavior across annotators with no dominant outlier.

For cases of disagreement between the internal annotators, we held discussions to reach a consensus on the final label. This resulted in our final data consist of 315 samples annotated with the scilence label.

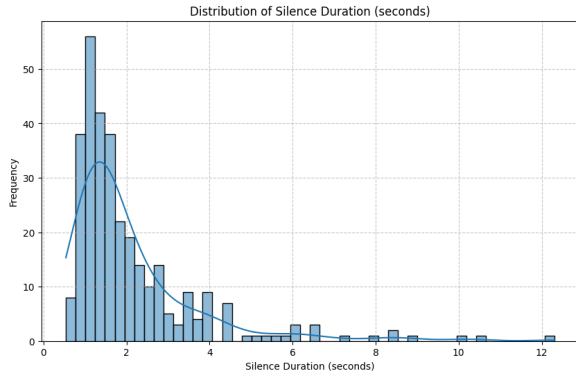


Figure 1: Distribution of silence durations (in seconds) within the dataset, showing a high frequency of short-duration intervals.

Label	Count	Percentage (%)
PRAGMATIC	108	34
RETRIEVAL	72	23
PROCESSING	71	22
EMOTIONAL	37	12
SURPRISE	18	6
EMBARRASSMENT	9	3
Total	315	100.0

Table 3: Distribution of silence labels in the annotated dataset.

3.7 Distribution of Silence Duration and Labels

The dataset exhibits a right-skewed distribution of silence durations, as illustrated in Fig. 1, with the majority of intervals concentrated between 1 and 2 seconds and a sparse tail extending up to 12 seconds. This physical distribution aligns with the label frequencies reported in Table 3, where over 75% of instances are categorized as PRAGMATIC, RETRIEVAL, or PROCESSING, functions typically associated with the brief pauses required for turn-taking and cognitive formulation. In contrast, the relative scarcity of emotionally-oriented or social silences suggests that while longer, more complex pauses occur, they represent distinct but rare pragmatic events in natural conversational podcasts.

4 The Annotation Task

As part of this project, we established an annotation task to categorize conversational silences and provide a robust dataset for analysis. Ten student annotators from the class labeled the dataset, with each student randomly assigned a subset of samples. This process ensured that each sample was annotated at least five times. This section describes the web-based annotation platform and analyzes

the results by comparing the external student annotations against the internal research team’s baseline.

4.1 Dedicated Web Platform

We developed a dedicated web-based annotation platform³ to support efficient and consistent labeling of silence segments. The interface presents each item as a three-part context: the transcript preceding the silence, a clearly marked We developed a custom web interface to ensure consistent silence labeling. The view presents the text context and silence duration alongside an embedded audio player. Annotators select one of six categories, with the option to add notes. The system ensures data integrity by timestamping submissions and allowing users to update their previous decisions.

Crucially, the platform captures detailed interaction metadata to trace user engagement. Beyond simply recording if the audio was played, we log the precise frequency of play, pause, and directional seek events. These actions are stored as a chronological sequence with timestamps, creating a complete history of the session up to the moment of submission. This granular data allows us to reconstruct the listening behavior for every item, enabling us to reliably distinguish between annotations made with active listening and those without a distinction central to the analysis in Section 4.2.3.

4.2 Annotation Task Analyses

4.2.1 Annotation Statistics Overview

Following the expansion of the annotation task to external participants, we collected a total of 1,425 individual labels across 315 unique audio segments. Each segment was annotated by at least five different individuals. The participant pool consisted of the four internal research team members and ten external student annotators from the class. Analysis of behavioral metadata from our custom interface revealed that in 95.5% of cases, annotators played the audio segment before assigning a label, indicating high task engagement and protocol compliance.

4.2.2 External vs. Internal Annotator Agreement

To evaluate differences in annotation reliability between groups, we calculated Cohen’s Kappa across various annotator subsets (see Figure 2). The internal research team achieved substantial agreement with $\kappa=0.70$ (77.7% raw agreement), establishing

³<https://talkleinbgu.github.io/silence-tagging/>

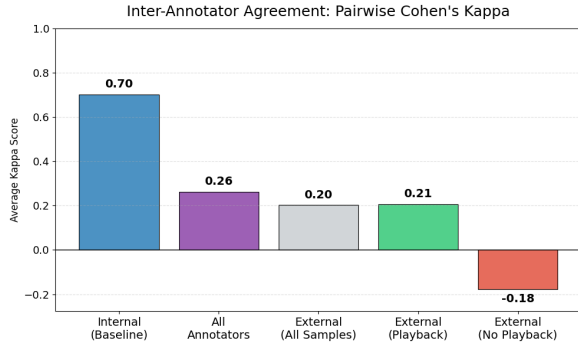


Figure 2: Inter-Annotator Agreement (Cohen’s κ) across annotator groups and conditions. Comparison includes internal team baseline, external annotators overall, internal-external majority vote agreement, and external annotators segmented by audio playback behavior.

a stable expert baseline. In contrast, when comparing the internal team’s majority vote with the external annotators’ majority vote, agreement dropped to $\kappa=0.26$ (40.6% raw agreement). Furthermore, agreement among external annotators themselves was lower at $\kappa=0.20$ (36.7% raw agreement). These findings demonstrate that silence categorization presents considerable complexity for annotators without extensive training, and that consistent labeling depends heavily on shared conceptual frameworks and clear category definitions.

4.2.3 The Listening Effect

Leveraging our platform’s audio playback tracking, we examined how listening behavior affected external annotator consistency. The green and the red bars in Fig. 2 show that when external annotators played the audio before labeling, their inter-annotator agreement was $\kappa=0.21$ (35.6% raw agreement). However, when audio was not played, agreement collapsed to $\kappa=-0.18$ (20.0% raw agreement), worse than chance. This "listening effect" provides strong evidence that auditory context is essential for silence interpretation. Prosodic features, intonation patterns, and acoustic cues appear critical for accurate categorization; without these signals, text-based decisions become effectively arbitrary.

4.2.4 Label Shift Patterns

Table 4 compares the internal team’s baseline with external annotators’ majority vote. PRAGMATIC silences showed highest stability (80.28% match), indicating this category is readily identifiable. EMOTIONAL silences achieved 72.97% agreement, but mismatches predominantly shifted to PROCESSING. This confusion is attributable to longer pause durations: external annotators in-

terpreted extended silences as cognitive deliberation rather than emotional processing, suggesting duration alone insufficiently differentiates these categories. The most pronounced divergences occurred with SURPRISE (50.00% match) and EMBARRASSMENT (55.56% match), where approximately half the instances were reclassified as PRAGMATIC. While the internal team identified these as emotionally-charged reactions, external annotators interpreted them as ordinary turn-transitions, suggesting that recognizing subtle emotional nuances requires deeper contextual engagement.

5 LLM Performance on Silence Classification

To evaluate how large language models interpret conversational silences, we tested Gemini-2.0-flash classification performance against our human-annotated ground truth labels. The model was presented with silence segments in two modalities: text-only transcripts with marked silence tokens, and text paired with corresponding audio clips. Figure 3 presents Cohen’s Kappa scores comparing Gemini’s predictions against ground truth across three conditions: without audio vs. ground truth ($\kappa = 0.0474$), with audio vs. ground truth ($\kappa = 0.1268$), and the model’s internal consistency between its own audio and non-audio predictions ($\kappa = 0.2982$). While audio provision marginally improved performance, all scores remained substantially below acceptable agreement thresholds, indicating that the model struggles fundamentally with this task. Notably, the highest agreement score emerged from comparing the model’s own predictions across modalities rather than against human labels, suggesting the model maintains internal consistency in its reasoning but diverges systematically from human interpretation.

Comparing LLM performance to human annotation patterns reveals a critical gap. While external human annotators achieve $\kappa = 0.26$ with the ground truth, the model’s agreement with ground truth remained approximately half this level even with audio ($\kappa = 0.1268$). This performance deficit likely stems from insufficient training data for silence categorization, current LLM corpora emphasize textual content while treating silence as formatting noise rather than meaningful communicative signals. The parallel between human and model performance regarding audio is noteworthy: both

Table 4: Comparison of internal team baseline labels against majority vote from all annotators. Match % indicates agreement rate; Majority Label of Unmatched shows the most common alternative label assigned by external annotators when disagreement occurred.

Label	Team Baseline	Majority Voting	Match Count	Match %	Majority Label of Unmatched
SURPRISE	18	18	9	50.00%	PRAGMATIC
EMBARRASSMENT	9	11	5	55.56%	PRAGMATIC
PROCESSING	108	95	68	62.96%	PRAGMATIC
EMOTIONAL	37	38	27	72.97%	PROCESSING
RETRIEVAL	72	65	46	63.89%	PROCESSING
PRAGMATIC	71	88	57	80.28%	EMOTIONAL

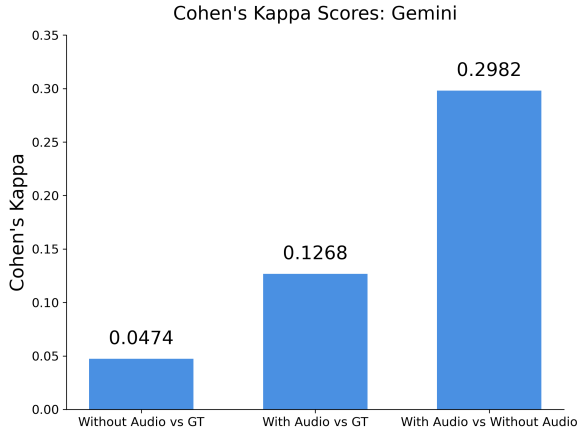


Figure 3: Cohen’s Kappa scores for Gemini LLM across different conditions: text-only vs. ground truth, with audio vs. ground truth, and internal consistency between modalities.

demonstrate improved consistency when auditory context is available, yet humans still substantially outperform the model. This suggests that while prosodic cues provide valuable information, interpreting silence functions requires pragmatic reasoning and conversational understanding that current models lack, underscoring the need for specialized training and architectures that better integrate multimodal signals with pragmatic knowledge.

6 Reflection

The process of constructing and executing this annotation task provided valuable insights into the challenges of capturing conversational phenomena through human annotation and automated classification, extending beyond the specific findings reported in previous sections.

The development of our custom annotation platform proved essential to the project’s success. The interface’s clean design, synchronized audio-transcript presentation, and progress tracking fea-

tures enabled annotators to work efficiently while maintaining engagement. The ability to pause and resume work at any point, combined with real-time transcription support, allowed annotators to follow dialogue accurately even when audio quality varied. These design decisions underscore the importance of investing in preprocessing and infrastructure before launching large-scale annotation efforts, careful preparation significantly streamlined data collection and improved annotation quality.

However, the task also revealed substantial challenges. Sourcing appropriate data proved more difficult than anticipated. We needed to balance multiple constraints: Hebrew-language content, two-speaker conversations with minimal background noise, and sufficient diversity in conversational styles. Technical hurdles included identifying reliable Hebrew transcription services and managing data access from various platforms. Perhaps most significantly, the absence of an "uncertain" option forced annotators to commit to categorical labels even when genuinely unsure. A more robust solution would involve incorporating explicit confidence ratings to weight or filter annotations based on annotator certainty before aggregation.

More broadly, our findings demonstrate that silence classification represents a genuinely difficult task requiring nuanced pragmatic and social reasoning. The substantial performance gap between trained internal annotators and external participants highlights how much implicit knowledge contributes to consistent interpretation. The even larger gap between human and LLM performance reveals fundamental limitations in current models’ ability to process non-linguistic communicative signals. Nevertheless, humans consistently outperform machines. This shows a superior ability to integrate contextual cues, prosodic information, and

conversational knowledge. This suggests that high-quality human annotations can serve as valuable training data. Such datasets provide a foundation for specialized models or fine-tuning existing architectures to better capture the pragmatic dimensions of human conversation.

Appendix

A Division of Work

Table 5 presents the distribution of tasks among team members throughout the project.

B Use of LLMs in the Project

Throughout this project, we utilized several large language models as assistive tools for different aspects of our work:

- **Documentation and Writing:** We used Gemini 3⁴ pro to assist with drafting and refining sections of the project report, improving clarity and structure of our written deliverables.
- **Annotation Platform Development:** The custom web-based annotation interface was developed with assistance from Claude Opus 4.5⁵ via GitHub Copilot, which provided code suggestions and helped streamline the implementation of audio playback tracking and behavioral metadata collection features.
- **Literature Review:** We employed NotebookLM⁶ to organize and synthesize related work, helping us identify relevant papers and extract key concepts from the existing literature on conversational silence and pragmatic analysis.

All LLM-generated content was reviewed, verified, and edited by team members to ensure accuracy and alignment with project goals.

References

- [1] B. Atal and L. Rabiner. 1976. [A pattern recognition approach to voiced-unvoiced-silence classification with applications to speech recognition](#). *IEEE Transactions on Acoustics, Speech, and Signal Processing*, 24(3):201–212.
 - [2] John R Dupuche. 2013. Silence and dialogue. *Gesher*, 4(4):52–54.
 - [3] Ángel López Gutiérrez and Juan José Arroyo Paniagua. 2024. [An exploration of silence in communication](#). *European Public & Social Innovation Review*, 9:1–18.
 - [4] Sahar Khademi. 2022. A review of silence in conversation: Discoursal perspective. *International Journal of Language and Translation Research*, 2(1):51–67.
 - [5] Dennis Kurzon. 2007. Towards a typology of silence. *Journal of pragmatics*, 39(10):1673–1688.
 - [6] Baoqin Wu, Muhammad Afzaal, and Dina Abdel Salam El-Dakhs. 2025. [‘yet his silence said volumes’: a pragmatic analysis of conversational silence in rapport management](#). *Cogent Arts & Humanities*, 12(1):2451490.
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- ⁴<https://blog.google/products-and-platforms/products/gemini/gemini-3/>
- ⁵<https://www.anthropic.com/news/claude-opus-4-5>
- ⁶<https://notebooklm.google/>

Task	Eliya	Ruben	Tal	Ziv
Data Collection	✓			✓
Annotation Platform Development			✓	
Related Work and Relevant Papers		✓		
Internal Annotation	✓	✓	✓	✓
Analyses	✓	✓	✓	✓
Report Writing	✓	✓	✓	✓

Table 5: Division of work among team members. Checkmarks indicate primary responsibility for each task.